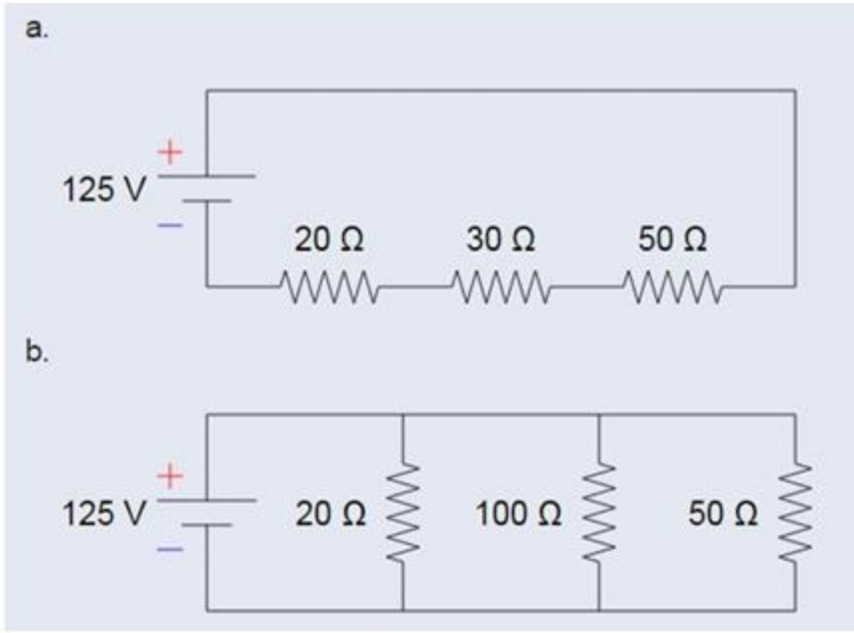
Practical Exercise 2: Circuit Design



Series Circuit

1. Calculate the total resistance in ohms,

$R_T = R_1 + R_2 + R$
$= 20 + 30 + 50$
$= 100\ \Omega$

2. Calculate the tot l current from the power supply

$I = V/R$
$= \frac{125}{100}$
$= 1.25\ A$

Current

3. Calculate the c rrent through each resistor

Since it is in series, is same

Current = 1.25 for all 3.

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4. Calculate the voltage drop through each resistor.

$V = IR$

for $20\,\Omega = 1.25 \times 20$

$= 25\,V$

$30\,\Omega = 1.25 \times 30$

$= 37.5\,V$

$50\,\Omega = 1.25 \times 30$

$= 62.5\,V$

5. Calculate the power dissipated through each resistor.

$P = VI$

For $20\,\Omega = 25 \times 1.25$

$= 31.25\,W$

$30\,\Omega = 37.5 \times 1.25$

$= 46.875\,W$

$50\,\Omega = 62.5 \times 1.25$

$= 78.125\,W$

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Parallel Circuit

1. Total resistance

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$= 0.08$$

$$R_T = \frac{1}{0.08}\ 12.5\,\Omega$$

$$=$$

2. Total current from power supply

$$I_T = \frac{V}{R_T}$$

$$= \frac{125}{12.5}$$

$$= 10\ A$$

3. Current through each resistor

$$\left.\begin{aligned} I &= V/R \\ &= 125/20 \\ &= 6.25\ A \end{aligned}\right\}\ \text{For } 20\,\Omega$$

For $100\,\Omega$ $= 125/100$

$= 1.25\ A$

$50\,\Omega$ $= 125/50 = 2.5\ A$

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4. Voltage drop through each resistor

Voltage is same across

$V = 125\,V$

5. Power dissipated through each resistor

$P = VI$

For $20\,\Omega$ $= 125 \times 6.25$

$= 781.25\,W$

$100\,\Omega$ $= 125 \times 1.25$

$= 156.25\,W$

$50\,\Omega$ $= 125 \times 2.5$

$= 312.5\,W$